

WELCOME

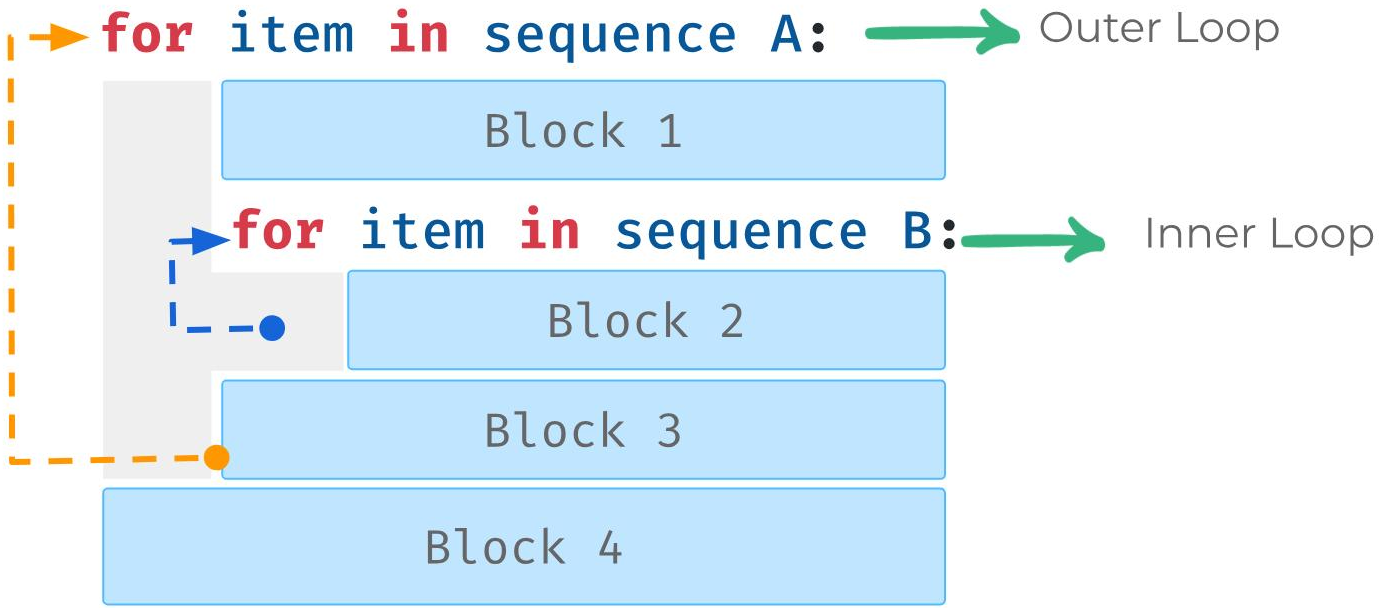
CCBP 4.0

21st Century Skills

Loop Control Statements

Recap

Nested Loops



Recap

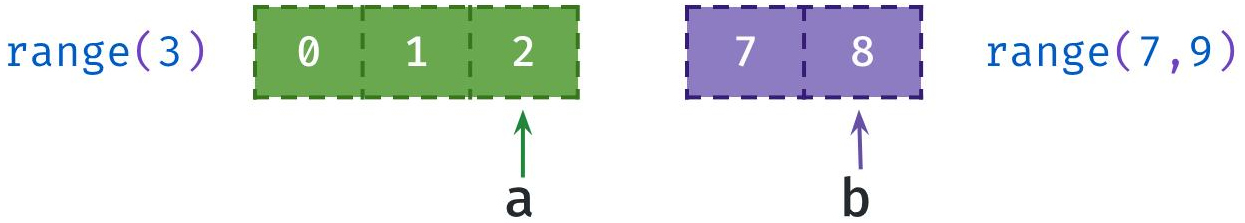
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

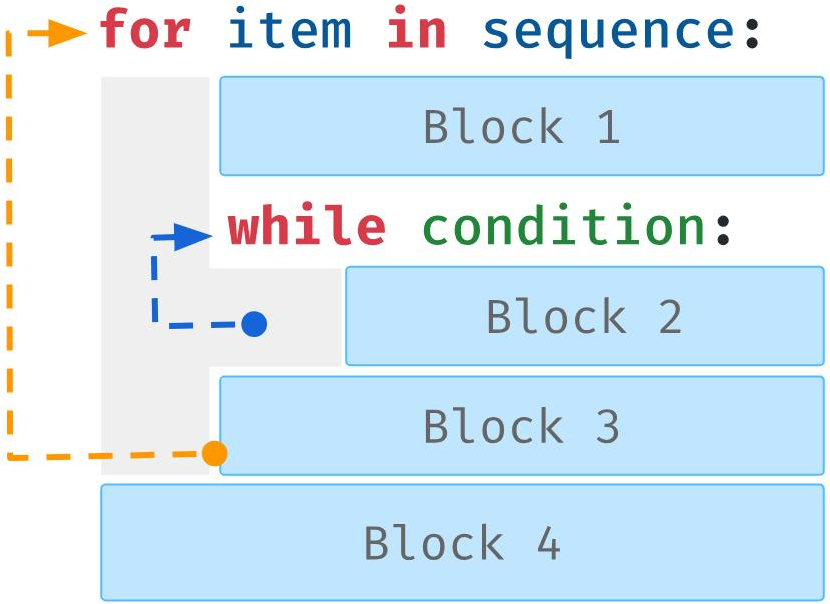
```
outer : 0
  inner : 7
  inner : 8
outer : 1
  inner : 7
  inner : 8
outer : 2
  inner : 7
  inner : 8
End
```



NXT WAVE

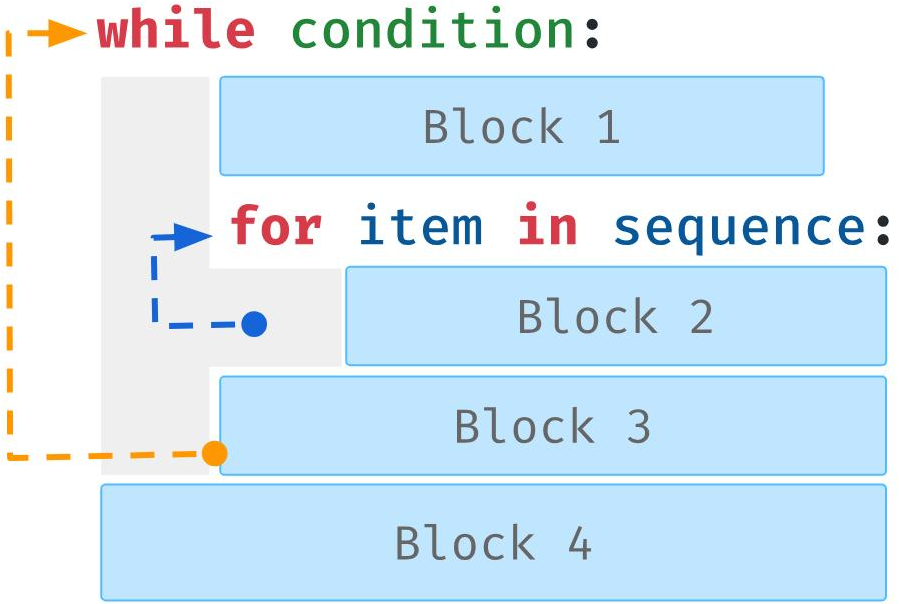
Recap

Nested Loops



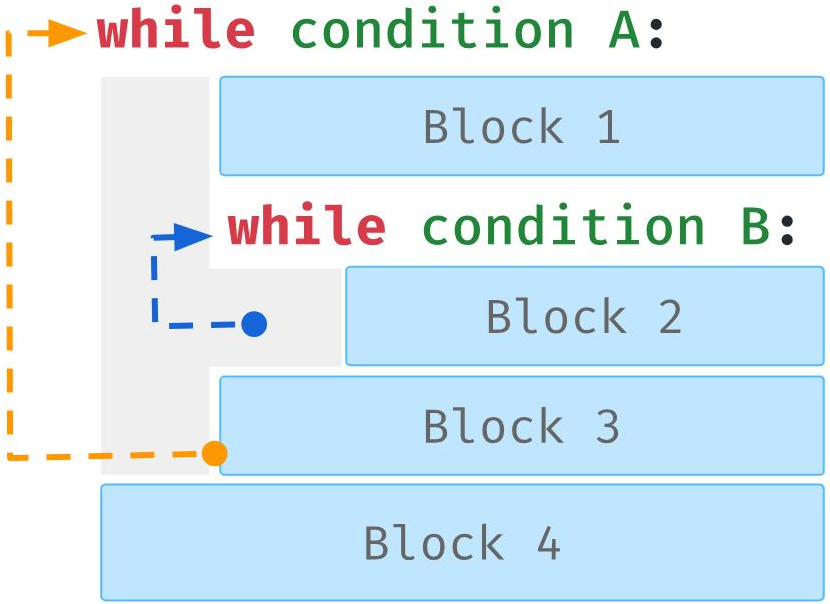
Recap

Nested Loops



Recap

Nested Loops



Daily Challenge

Identify The Mistake

Code

```
n = int(input())
for row_num in range(0,n+1):
    row_output = ""
    seq_num = row_num
    while seq_num > 0:
        row_output = row_output + str(seq_num)
        seq_num = seq_num - 1
    print(row_output)
```

Input

```
3
```

Expected Output

```
321
21
1
```

NXT WAVE

Daily Challenge

Identify The Mistake

Code

```
n = int(input())
for row_num in range(0,n+1):
    row_output = ""
    seq_num = row_num
    while seq_num > 0:
        row_output = row_output + str(seq_num)
        seq_num = seq_num - 1
    print(row_output)
```

Input

```
3
```

Output

```
1
21
321
```

NXT WAVE

Daily Challenge

Identify The Mistake

Code

```
n = int(input())
for row_num in range(0,n+1):
    row_output = ""
    seq_num = n - row_num
    while seq_num > 0:
        row_output = row_output + str(seq_num)
        seq_num = seq_num - 1
    print(row_output)
```

Input

```
3
```

Output

```
321
21
1
```

NXT WAVE

Daily Challenge

Identify The Mistake

Code

```
n = int(input())
for row_num in range(0,n):
    row_output = ""
    seq_num = n - row_num
    while seq_num > 0:
        row_output = row_output + str(seq_num)
        seq_num = seq_num - 1
    print(row_output)
```

Input

```
3
```

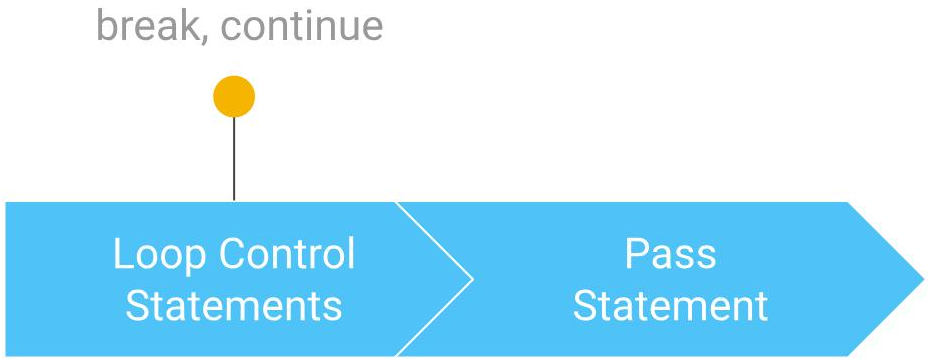
Output

```
321
21
1
```

NXT WAVE

Agenda for Today's Session

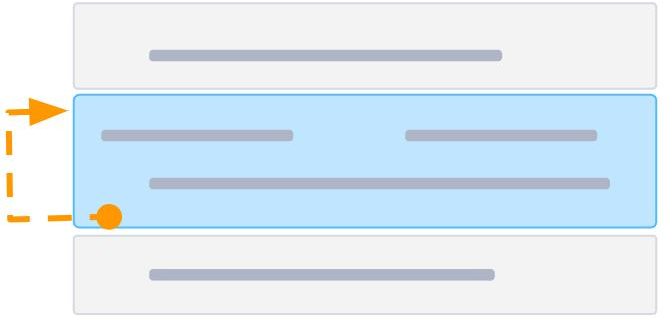
Loop Control Statements



Scenario

Loops

Loops allow us to execute
a block of code several times



Control Statements

Alter the
sequential execution
of a program

→ `if-elif-else`

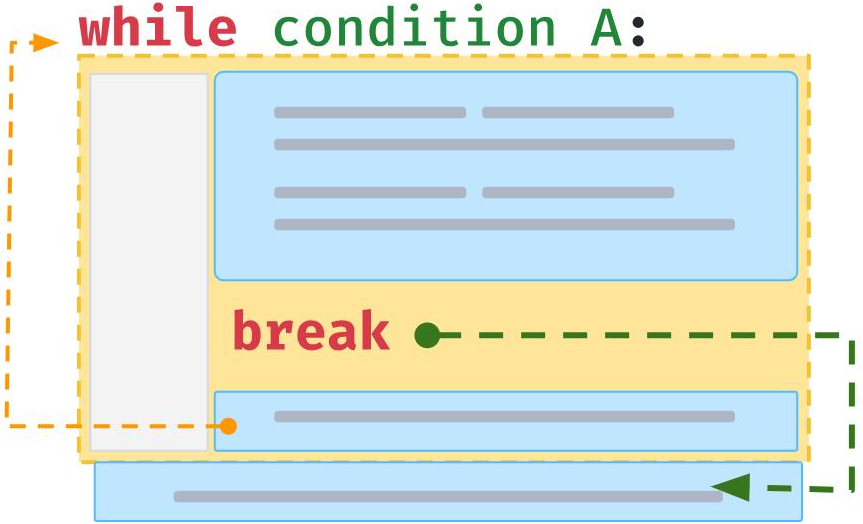
→ `while, for`

→ `break, continue`

Break

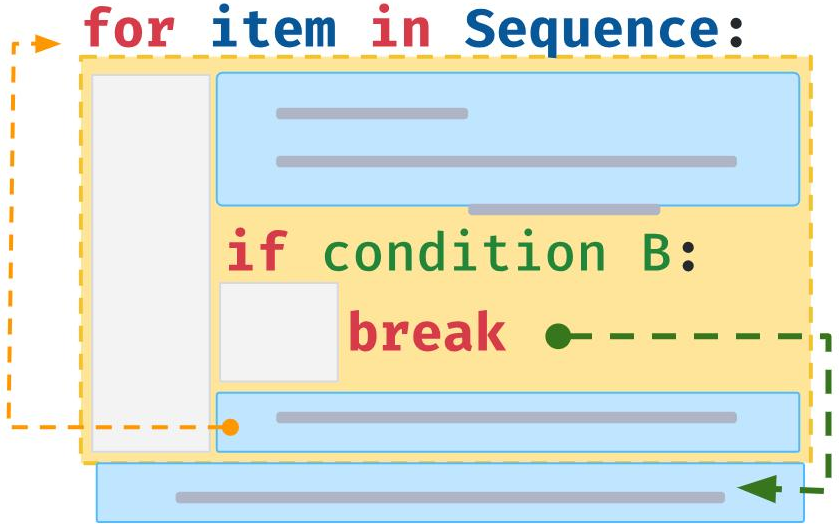
Break

Break makes the program **exit a loop** early



Break

Generally, break is used
to exit a loop
when a condition is satisfied



Loop Control Statements

Break

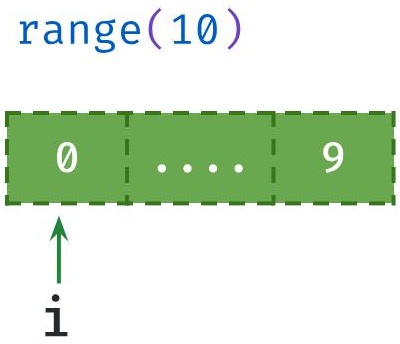
Code

```
for i in range(10):
    if i == 2:
        break
    print(i)
print("End")
```

Break

Code

```
for i in range(10):
    if i == 2:
        break
    print(i)
print("End")
```

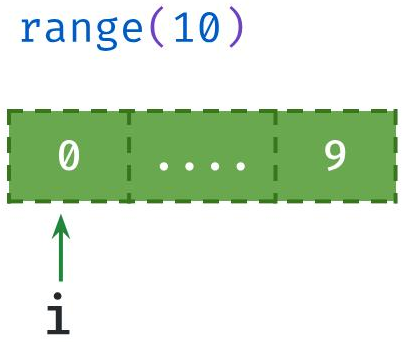


Break

Code

```
for i in range(10):
    if i == 2:
        break
    print(i)
print("End")
```

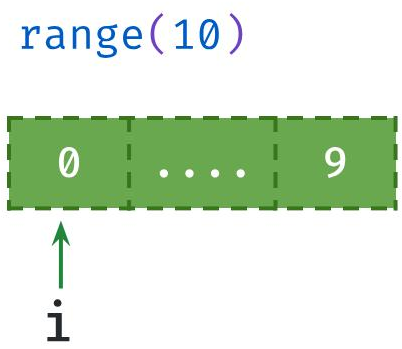
False



Break

Code

```
for i in range(10):
    if i == 2:
        break
    print(i)
print("End")
```



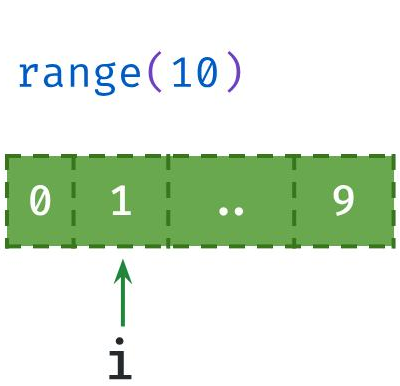
Output

```
0
```

Break

Code

```
for i in range(10):
    if i == 2:
        break
    print(i)
print("End")
```



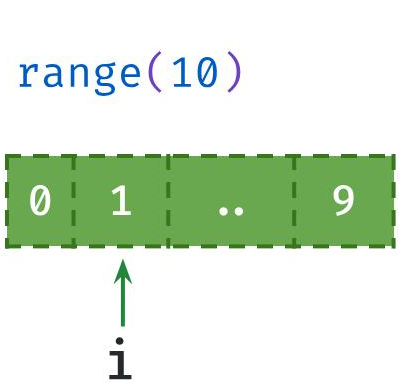
Output

```
0
```

Break

Code

```
for i in range(10):
    if i == 2:          False
        break
    print(i)
print("End")
```



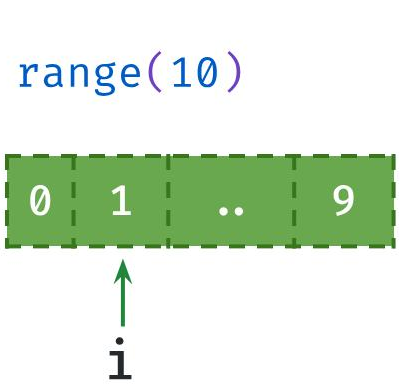
Output

```
0
```

Break

Code

```
for i in range(10):
    if i == 2:
        break
    print(i)
print("End")
```



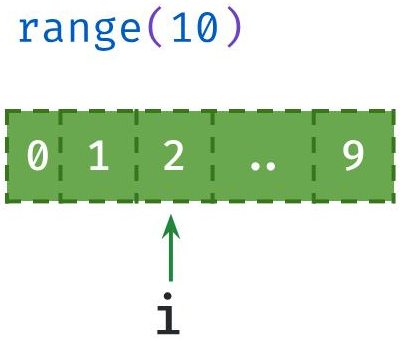
Output

```
0
1
```

Break

Code

```
for i in range(10):
    if i == 2:
        break
    print(i)
print("End")
```



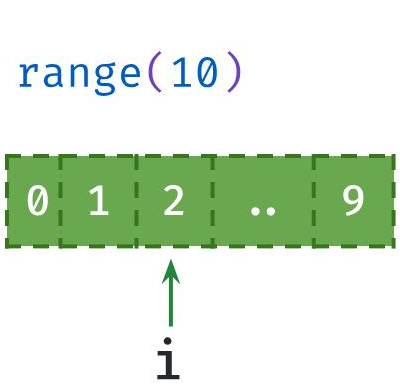
Output

```
0
1
```

Break

Code

```
for i in range(10):
    if i == 2:          True
        break
    print(i)
print("End")
```



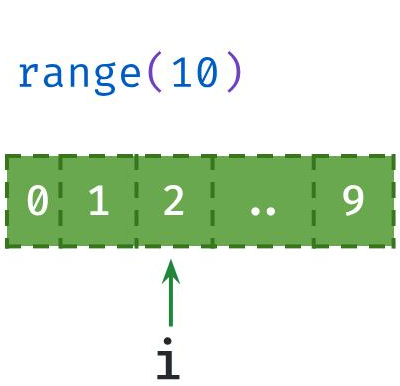
Output

```
0
1
```

Break

Code

```
for i in range(10):
    if i == 2:
        break
    print(i)
print("End")
```



Output

```
0
1
```

Break

Code

```
for i in range(10):
    if i == 2:
        break
    print(i)
print("End")
```

Output

```
0
1
End
```

Loop Control Statements

Break

Code

```
i = 0
while i < 10:
    if i == 2:
        break
    print(i)
    i = i + 1
print("End")
```

Loop Control Statements

Break

Code

```
i = 0
while i < 10:
    if i == 2:
        break
    print(i)
    i = i + 1
print("End")
```

Output

```
0
1
End
```

Break

Code

```
i = 0
while i < 10:
    break
    print(i)
    i = i + 1
print("End")
```

What will be the output of the given code?

Break

Code

```
i = 0
while i < 10:
    break
    print(i)
    i = i + 1
print("End")
```

Output

```
End
```

Break

Code

```
for i in range(10):
    print(i)
break
print("End")
```

What will be the output of the given code?

Loop Control Statements

Break in Python

Code

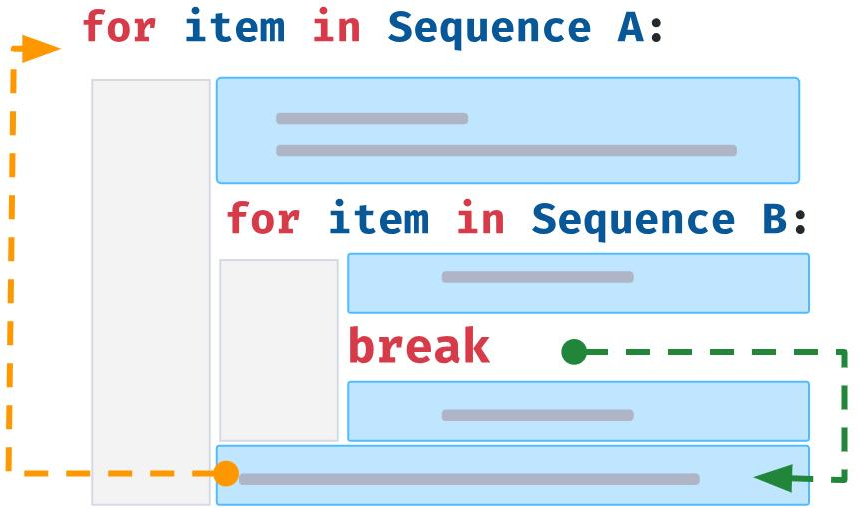
```
for i in range(10):
    print(i)
break
print("End")
```

Output

```
SyntaxError:
'break' outside loop
```

Break in Nested Loop

Break in inner loop stops the execution of the inner loop



Break in Nested Loop

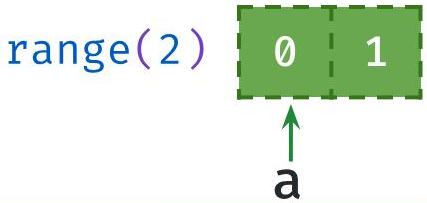
Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```



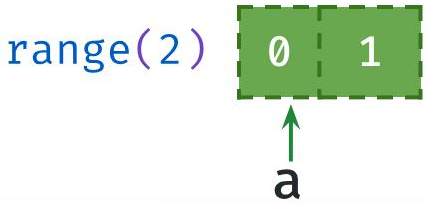
Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
```



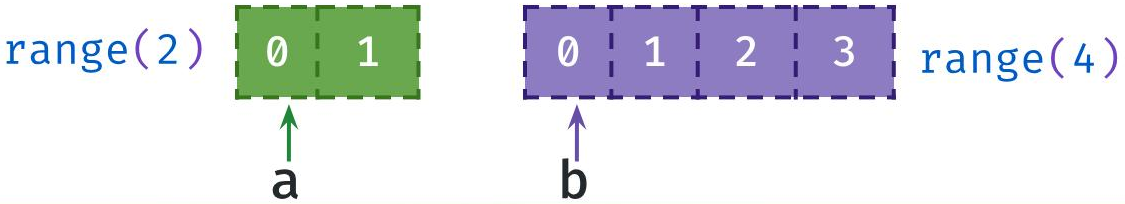
Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
```



Break in Nested Loop

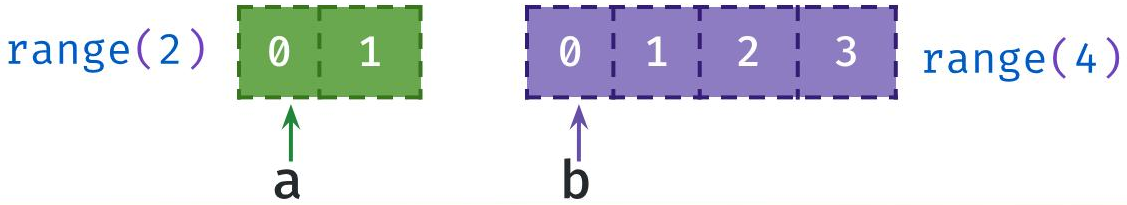
Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

False

Output

```
outer : 0
```



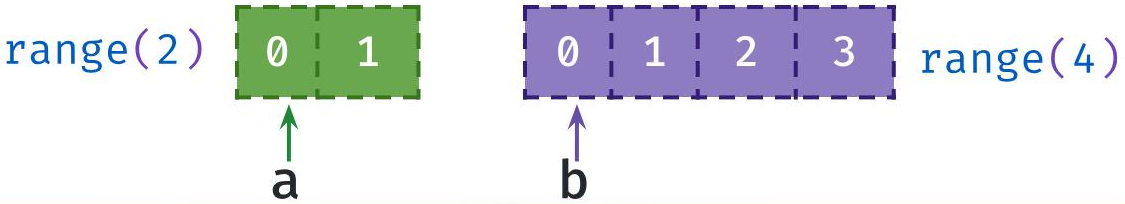
Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
 inner : 0
```



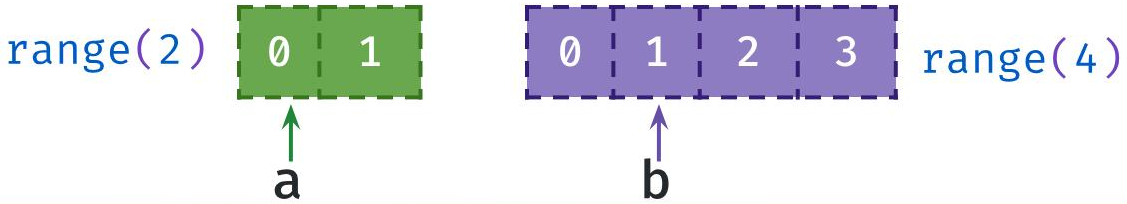
Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
 inner : 0
```



Break in Nested Loop

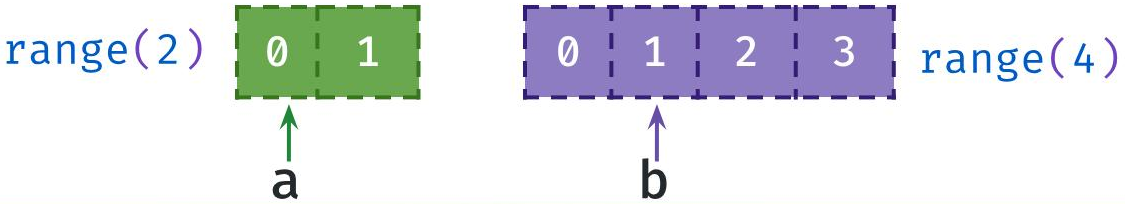
Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

True

Output

```
outer : 0
 inner : 0
```



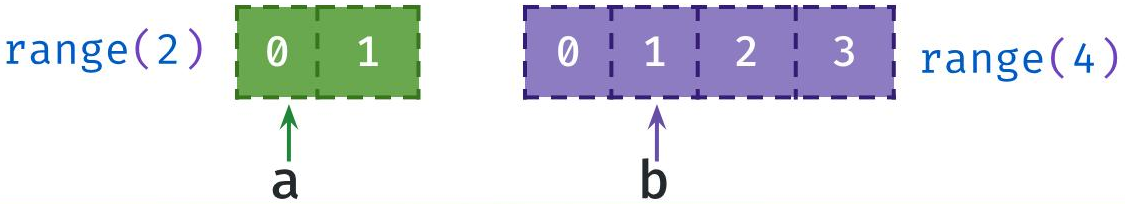
Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
 inner : 0
```



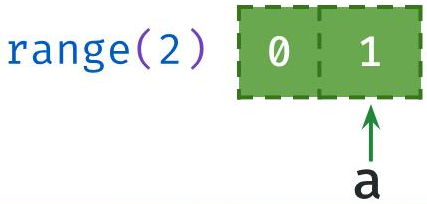
Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
 inner : 0
```



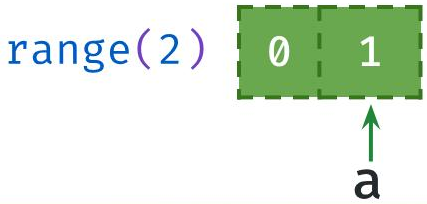
Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
 inner : 0
outer : 1
```



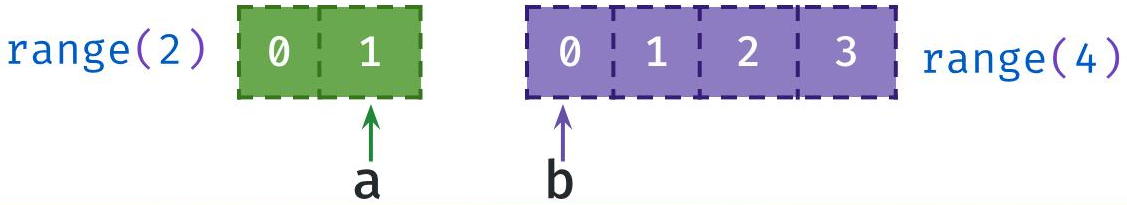
Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
 inner : 0
outer : 1
```



Break in Nested Loop

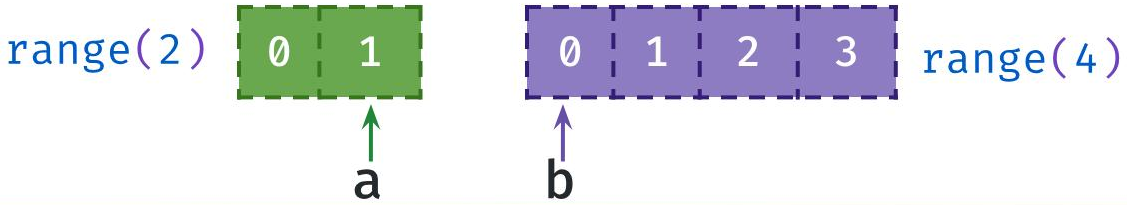
Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

False

Output

```
outer : 0
 inner : 0
outer : 1
```



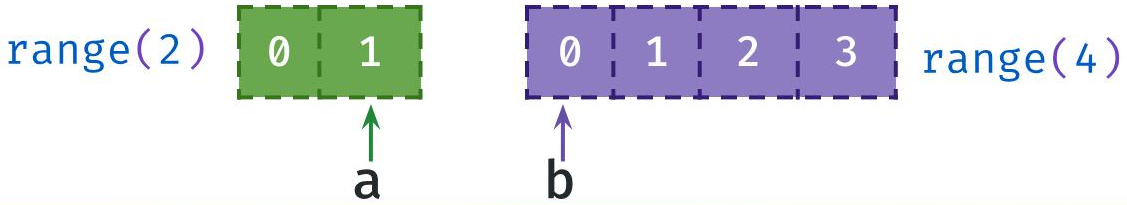
Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
 inner : 0
outer : 1
 inner : 0
```



NXT WAVE

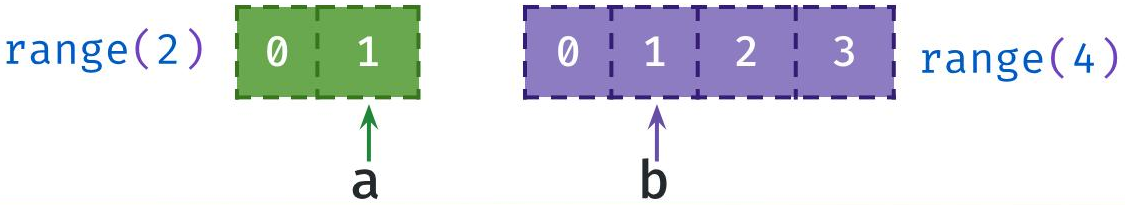
Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
 inner : 0
outer : 1
 inner : 0
```



Break in Nested Loop

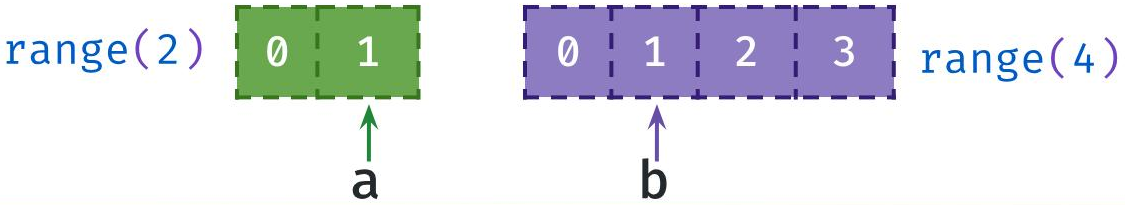
Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

True

Output

```
outer : 0
 inner : 0
outer : 1
 inner : 0
```



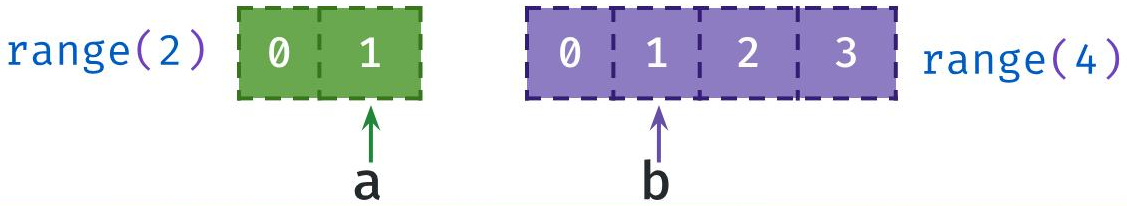
Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
 inner : 0
outer : 1
 inner : 0
```



Break in Nested Loop

Code

```
for a in range(2):
    print("outer : "+ str(a))
    for b in range(4):
        if b == 1:
            break
        print(" inner : "+ str(b))
```

Output

```
outer : 0
 inner : 0
outer : 1
 inner : 0
```

Question

First Word in Uppercase

Given a sentence,

print a new sentence

with the first word in uppercase.

Input

```
Python is a programming language.
```

Output

```
PYTHON is a programming language.
```

Question

Input / Output Format

Input Format

Input will contain a string (sentence)

Input

```
Line of code
```

Output

```
LINE of code
```

Question

Input / Output Format

Output Format

Print a new sentence with the first word in uppercase

Input

```
Python has a simple syntax.
```

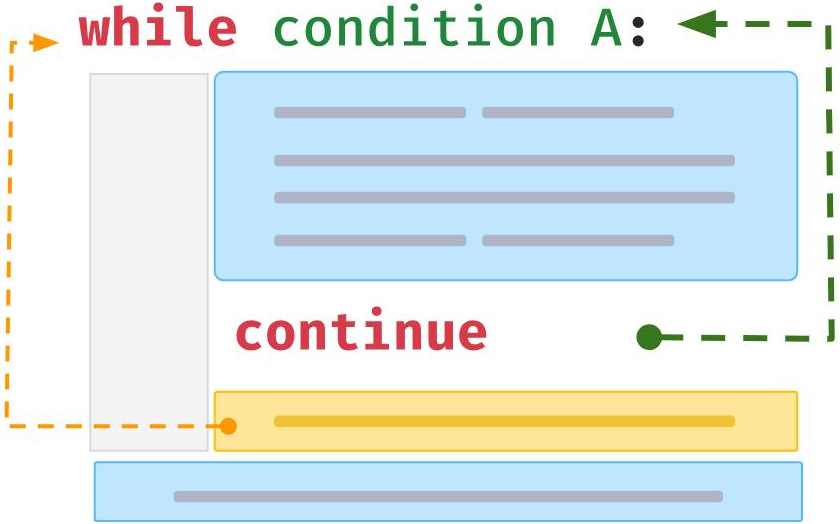
Output

```
PYTHON has a simple syntax.
```

Continue

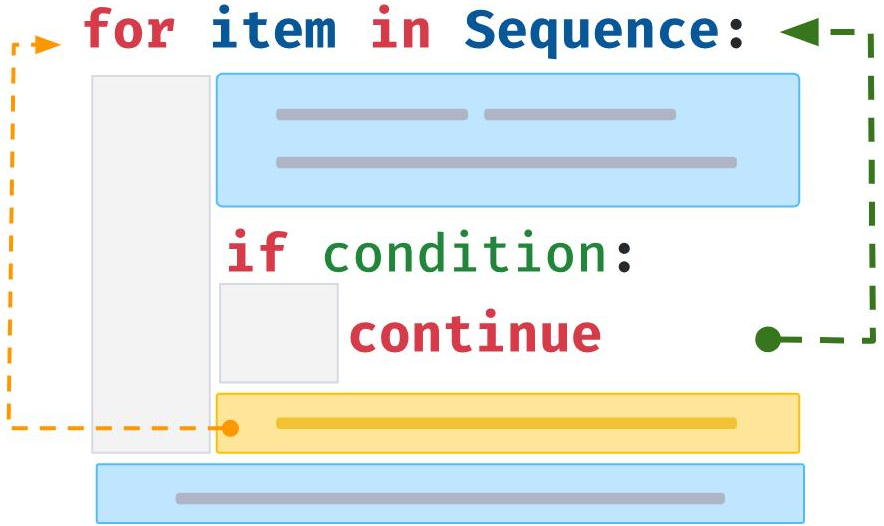
Continue

Continue makes the program skip the remaining statements in the current iteration and begin the next iteration



Continue

Generally, continue is used to skip the remaining statements in the current iteration when a condition is satisfied



Loop Control Statements

Continue

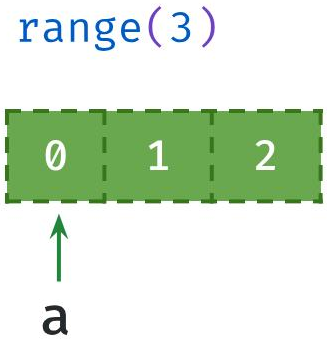
Code

```
for a in range(3):
    if a == 1:
        continue
    print(a)
print("End")
```

Continue

Code

```
for a in range(3):
    if a == 1:
        continue
    print(a)
print("End")
```

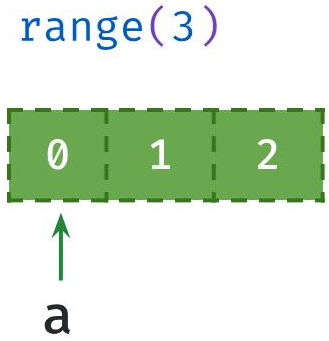


Continue

Code

```
for a in range(3):
    if a == 1:
        continue
    print(a)
print("End")
```

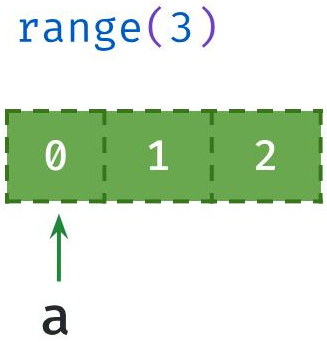
False



Continue

Code

```
for a in range(3):
    if a == 1:
        continue
    print(a)
print("End")
```



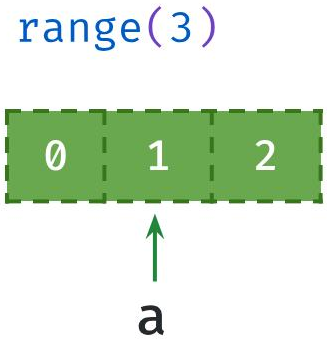
Output

```
0
```

Continue

Code

```
for a in range(3):
    if a == 1:
        continue
    print(a)
print("End")
```



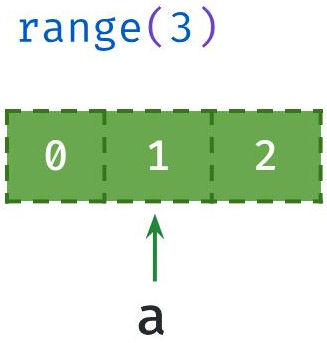
Output

```
0
```

Continue

Code

```
for a in range(3):
    if a == 1:          True
        continue
    print(a)
print("End")
```



Output

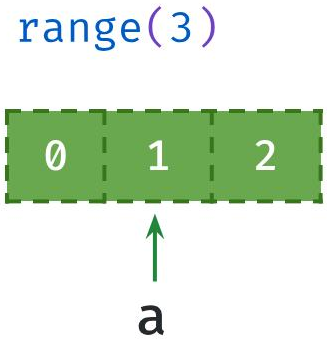
```
0
```

Loop Control Statements

Continue

Code

```
for a in range(3):
    if a == 1:
        continue
    print(a)
print("End")
```



Output

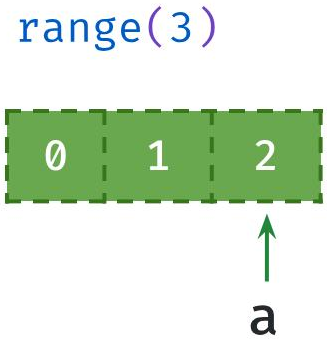
```
0
```

Loop Control Statements

Continue

Code

```
for a in range(3):
    if a == 1:
        continue
    print(a)
print("End")
```



Output

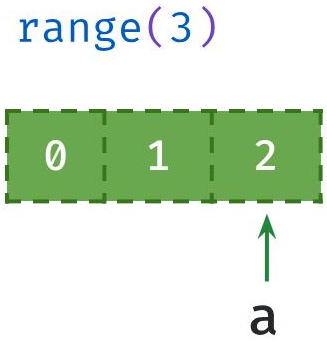
```
0
```

Continue

Code

```
for a in range(3):
    if a == 1:
        continue
    print(a)
print("End")
```

False



Output

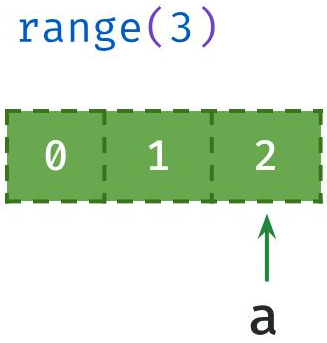
```
0
```

Loop Control Statements

Continue

Code

```
for a in range(3):
    if a == 1:
        continue
    print(a)
print("End")
```



Output

```
0
2
```

Continue

Code

```
for a in range(3):
    if a == 1:
        continue
    print(a)
print("End")
```

Output

```
0
2
End
```

Continue

Code

```
for a in range(3):
    continue
    print(a)
print("End")
```

What will be the output of the given code?

Continue

Code

```
for a in range(3):
    continue
    print(a)
print("End")
```

Output

```
End
```

Continue

Code

```
for a in range(3):
    print(a)
continue
print("End")
```

What will be the output of the given code?

Loop Control Statements

Continue in Python

Code

```
for a in range(3):
    print(a)
continue
print("End")
```

Output

```
SyntaxError:
'continue' not
properly in loop
```

Pass

Empty Blocks

Pass

Pass statement is used as a syntactic placeholder.

When it is executed, nothing happens

Generally used when
we have to test the code
before writing the complete code

```
if condition A:
    Block 1
elif condition B:
    pass
else:
    Block 3
```

NXT WAVE

Pass

Code

```
age = int(input())
output = ""
if age >= 20:
elif age > 12:
else:
print(output)
```

Let's say, we are writing a code to print the growth stage of a person based on their age.

We want to test the code written so far.

Pass

Code

```
age = int(input())
output = ""
if age >= 20:
elif age > 12:
else:
print(output)
```



You cannot leave empty blocks

Pass

Code

```
age = int(input())
output = ""
if age >= 20:
    pass
elif age > 12:
    pass
else:
    pass
print(output)
```

Pass

Code

```
age = int(input())
output = ""
if age >= 20:
    output = "Your age is " + str(age) + ". You're an adult now."
elif age > 12:
    pass
else:
    pass
print(output)
```

Input

```
21
```

Output

```
Your age is 21. You're an adult now.
```

Pass

Code

```
age = int(input())
output = ""
if age >= 20:
    output = "Your age is " + str(age) + ". You're an adult now."
elif age > 12:
    output = "Your age is " + str(age) + ". You're a teenager now."
else:
    pass
print(output)
```

Input

```
14
```

Output

```
Your age is 14. You're a teenager now.
```

Pass

Code

```
age = int(input())
output = ""
if age >= 20:
    output = "Your age is " + str(age) + ". You're an adult now."
elif age > 12:
    output = "Your age is " + str(age) + ". You're a teenager now."
else:
    output = "Your age is " + str(age) + ". You're a child now."
print(output)
```

Input

```
8
```

Output

```
Your age is 8. You're a child now.
```

Empty Loops

```
while condition A:
    pass
```

```
for item in Sequence:
    pass
```

Pass

Code

```
n = int(input())
for i in range(n):
    if i == 3:
        pass
    print(i)
```

Input

```
5
```

Output

```
0
1
2
3
4
```

Best Practices

Key Takeaways

- Loop Control Statements
  - Break
  - Continue
- Pass
  - Empty Conditions
  - Empty Loops



Next Session

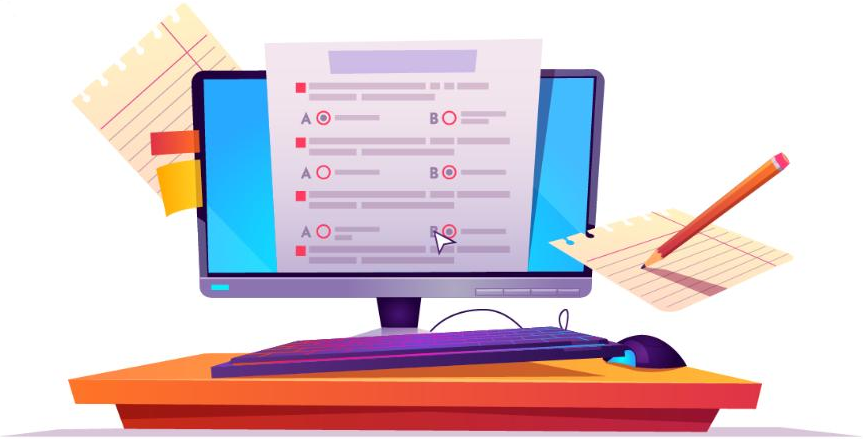
Rounding Numbers

Naming Variables

Comparing Strings

and many more ...

Practice



Daily Challenge

Identify The Mistake

Code

```
n = int(input())
is_prime = True
for i in range(2, n):
    if (i % n) == 0:
        is_prime = False
        break
print(is_prime)
```

Check if the given number is a prime number

Input

```
22
```

Expected Output

```
False
```

NXT WAVE

NXT WAVE

THANK YOU



ALL THE BEST